LAB 02 - PROMPTS

Data-Validation Stage: Cross-Check Pipeline Bars against TradingView

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab02-tradingview-charts/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Add the data-check panel to SGX:D05 (A3, K9)

Prompt - copy into your AI coding assistant
Explain labs_fullcode/Lab02-tradingview-charts/lcc_data_check.pine for a finance professional: what each input does, how the script finds the bar to validate, and which values in its table I should copy into tradingview_values.csv. Do not change the file.

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Check: The orange-shaded bar is 2025-12-31 and the panel shows about open 56.50, high 56.60, low 56.11, close 56.36, volume about 2.05 million (SGD, default unadjusted view). Free plans may show SGX data delayed.

Step 2: Repeat for NASDAQ:AAPL and switch the adjustment (A3, K9)

Do this (no prompt needed)
Open NASDAQ:AAPL on 1D with the same indicator and read 31 December 2025. Then open Chart settings > Symbol and tick 'Adjust data for dividends'; read the bar again and untick it.

Check: Default view: open 273.06, high 273.68, low 271.75, close 271.86, volume about 27.29 million. With 'Adjust data for dividends' ticked the close drops to about 271.12.

Step 3: Record what TradingView shows (A3)

Do this (no prompt needed)
Open Lab02-tradingview-charts/tradingview_values.csv. Overwrite the two example rows with your own readings (tradingview_symbol, date, open, high, low, close, volume, setting = default or adjusted).

Check: Two rows: NASDAQ:AAPL and SGX:D05, both dated 2025-12-31, setting default.

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Step 4: Build the validation stage (A3, K5, K9)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/validate.py. reference_bars(symbol) reads pipeline/reference/<SYMBOL>_1Day_raw.csv (columns timestamp, open, high, low, close, adj_close, volume, dividend, split). adjusted_bars(symbol, raw) returns the research bars from common.data.get_bars for the same dates and, if no file exists (FileNotFoundError, e.g. D05.SI), raw open/high/low/close multiplied by adj_close / close. compare(tv_row, bar) returns the percentage difference per field and match=True if every price is within 0.1% and volume within 10%. run(date='2025-12-31', tv_file=None) loops over config universe.watch, prints for each symbol the RAW row, the ADJUSTED row, your TradingView row from Lab02-tradingview-charts/tradingview_values.csv (lines starting with # are comments) and the differences, the dividend-adjustment factor (adjusted close / raw close), change vs prior close, 52-week high/low and 20-day average volume, and a verdict: 'PASS - TradingView matches RAW', 'PASS - TradingView matches ADJUSTED' or 'FAIL'. Save _outputs/Lab02_validation_report.csv and a chart Lab02_validation.png (raw vs adjusted close in 2025, one panel per symbol) with common.plotting.save. main(argv) supports --date and --tv. This stage only reads data - it must never call an order API. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: pipeline/validate.py exists and python -m pipeline validate --help lists --date and --tv.

Step 5: Run the validation stage (A3, K9)

Do this (no prompt needed)
Run: python -m pipeline validate --date 2025-12-31 (same as: python3 Lab02-tradingview-charts/validate_stage.py --date 2025-12-31).

Check: AAPL RAW 273.06 / 273.68 / 271.75 / 271.86 vs ADJUSTED 272.32 / 272.94 / 271.01 / 271.12, factor 0.9973, diff vs adjusted +0.272%. D05 RAW 56.50 / 56.60 / 56.11 / 56.36 vs ADJUSTED 54.35 / 54.45 / 53.98 / 54.22, factor 0.9620, diff +3.950%. Both 'PASS - TradingView matches RAW (dividend-unadjusted) bars'; RESULT: 2/2; _outputs/Lab02_validation_report.csv and Lab02_validation.png.

Step 6: Understand adjusted versus unadjusted prices (K9)

Prompt - copy into your AI coding assistant
Using the dividend and split columns in labs_fullcode/pipeline/reference/D05.SI_1Day_raw.csv, explain why the pipeline's adjusted D05 close for 2025-12-31 is about 3.95% below TradingView's close, why backtests must use adjusted prices while orders are priced in raw prices, and what error a trading program makes if it mixes the two. Answer in five bullet points; do not change any file.

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Check: D05's factor 0.9620 reflects dividends paid after the bar (back-adjusted into history); backtests use adjusted returns, orders use the raw market price.

Step 7: Make the validation fail on purpose (A3)

Do this (no prompt needed)
In tradingview_values.csv change the AAPL row to the dividend-adjusted values open 272.32, high 272.94, low 271.01, close 271.12, setting adjusted, and rerun the stage. Next change only the close to 280.00 and rerun. Finally restore your real reading.

Check: With 271.12 AAPL shows 'PASS - TradingView matches ADJUSTED bars'; with 280.00 it shows 'FAIL - TradingView matches neither series' and RESULT: 1/2.

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Step 8: Identify other forms of market data (K9)

Do this (no prompt needed)
On TradingView open the News, Financials and economic calendar views for SGX:D05. Note whether the SGX data are real-time or delayed on your plan and why the pipeline takes its US data from the Alpaca API rather than from a chart.

Check: You can name at least three data forms besides bars (news, fundamentals, economic calendar) and state whether your SGX data are delayed.

Run it

Step	Command / file
Run Lab 02	python3 Lab02-tradingview-charts/validate_stage.py (same as: python -m pipeline validate)
The complete code	pipeline/validate.py (reference copy: Lab02-tradingview-charts/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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